

So, Who Likes You?

Evidence from a Randomized Field Experiment

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Abstract

With one-third of marriages in the United States beginning online, online dating platforms have become important curators of the modern social fabric. Prior work on online dating has elicited two critical frictions in the heterosexual dating market: women, governed by age-old social norms of not making the first move, are inhibited in their interactions, in that they do not initiate contact with men. On the other side, men send an abundance of messages, the majority of which do not convert to matches. A key distinguishing feature of online dating versus its traditional counterpart is the ability to leave a range of digital signals not replicable in the offline world. These digital signals can impact the nature of online dating platform outcomes. In this paper, we study the impact of a feature that reveals ‘*who likes you*’ (WLY) on engagement, number of matches, match efficiency, and match sorting in online dating. This feature reveals the identity of the voters who have rated the focal user with a like. To causally identify the effect of this feature we conduct a large-scale randomized control trial in collaboration with a major North American dating platform. The treatment causes women to be more proactive, sending 7.4% more messages, which is a highly desirable market improvement given that men send double the number of messages as compared to women. Further, we find that the women endowed with this feature increase their matches by 14.4%, while men increase their matches by 11.5%. Analyzing the moderating impact of desirability—a key aspect of the WLY feature—provides us with nuanced findings. Depending on the levels of each of the two parties’ desirability, we see evidence of sorting, encouragement, and discouragement.

Keywords: online dating; IT-enabled features; identity revelation; digital signals; randomized trial; field experiment; matching markets; desirability

1. Introduction

Recent research has shown that one-third of marriages in the United States begin online (Cacioppo et al. 2013). Further, close to half of single people have gone on a date with someone they met online (Canaday and Ross 2017). Not surprisingly, the number of online dating platforms is large and growing, with estimates of approximately 2500 online dating platforms in the US alone (Zwilling 2013) and 1400 in the UK (Russell 2020). These platforms vary in their target audience (e.g. based on sexuality, religion, age), their objectives (e.g. long-term vs. short-term relationships), and an important aspect related to this study – the IT-enabled features they offer.

Digital platforms offer IT-enabled features to enhance users’ experiences. However, two-thirds of the features that platforms introduce do not significantly impact user behavior (Kohavi and Thomke 2017). Dating platforms, in particular, leverage features that allow users to leave a range of digital signals not replicable in the offline world. If done right, these could mitigate offline social frictions, and facilitate matching. However, given the inherent complexity of human behavior, especially in matters of the heart, it is unclear if these features are actually effective in achieving desirable outcomes and reducing key frictions in online dating platforms.

Prior work looking at the impact of IT-enabled features in the online dating context examines the *anonymous browsing* feature using a randomized field experiment, and finds that weak signaling is critical in achieving successful matches, particularly for women (Bapna et al. 2016). Weak signaling occurs when a non-anonymous profile view leaves a digital footprint (the weak signal) on another user’s profile, and often triggers a message in response to kickstart the matching process. In this study, we look at another IT-enabled feature, the *who likes you* (WLY) feature. We study the impact of this feature on engagement, matching, and match efficiency outcomes paying close attention to the heterogeneous effect of the treatment conditional on the

desirability of the two parties involved in the matching. We use the term desirability to encompass physical desirability expressed through the profile image of a user, but also other profile information that could reflect other characteristics like style, inferred personality, and inferred demeanor.

Online dating platforms typically allow users to “like” other users by providing the binary option to “like” or “pass” on a potential match or to “swipe right” or “swipe left” on a mobile device to give a positive or negative signal. Information from this liking process is used behind-the-scenes by the dating platforms for recommendations of potential matches. For the remainder of this paper we call the action of “liking” or “swiping right” on a potential mate a “like.” As you can see in Figure 1, the default setting is that the WLY feature is off, i.e. that the *identity* of the users who “like” is kept secret (see “Control” panel in Figure 1). The focal user (who, in this study is always the user who receives the like) can see that “X” number of users have liked them, but they cannot see the identity of the users who have liked them. Thus, the default setting in online dating resembles, in its own unique way, that of the offline dating market. There exists information asymmetry about people’s preferences for each other. The WLY feature specifically allows the focal user to observe the *identity* of the user who has (previously secretly) liked them by unblurring the image and revealing the user’s first name and age.² As a result it, arguably, breaks down the information asymmetry prevalent in the default scenario by revealing the profile picture of the user who sent the like to the receiver of the like. We believe, and our results show, that this reduction of information asymmetry should lead to a host of beneficial outcomes for participants.

²The focal user can further click on the image to check the targeting user’s profile page.

Figure 1. Who Likes You? Feature



We study the WLY feature’s impact on multiple online dating outcomes, including engagement, number of matches, and match efficiency (Bapna et al. 2016). Further, we go beyond prior work in examining sorting based on desirability in matching (Hitsch et al. 2010, Kalmijn 1998, Taylor et al. 2011). We posit that the effect of the WLY signal will be driven by the desirability of the users who are sending likes (targeting user) and receiving likes (focal user) from one another. Arguably, viewing the picture of the targeting users, alongside having the capability to access the targeting users’ profile for further information, allows the focal user to instinctively judge how desirable the targeting users are.³ This is the first-order difference between the treatment group and control group. In online dating, research has shown that a user’s profile picture in itself is the key aspect of his or her profile that drives choices. A Smithsonian Magazine⁴ article citing McGloin and Denes (2018) quote the author, “you look at a picture of someone and all of a sudden you're making judgments about what their personality is like, what their values are, whether or you

³The profile pictures are hyperlink enabled. Thus, by clicking on the profile picture, an user can visit the profile page linked with that profile picture.

⁴ <https://www.smithsonianmag.com/science-nature/your-online-dating-profile-picture-affects-whether-people-trust-you-180955207/>

want to go on a date with them or even maybe spend the rest of your life with them...and it's all based on one picture.” Further, Fiore et al. (2008) confirm prior research suggesting that “the attractiveness and other qualities of the photograph were the strongest predictors of whole profile attractiveness.” Given the hedonic nature of online dating, this is a critical difference we expect to dig deeper into and understand. Is there heterogeneity in the WLY treatment as a function of the desirability of the focal user and the targeting users? To the best of our knowledge prior research has not considered the importance of desirability revelation in the online dating context.

Therefore, based on the directives from extant literature and motivation from the nature of our treatment, we study engagement outcomes, matching outcomes, i.e. the number of matches, match efficiency, and the nature of matches, i.e. the nature of desirability sorting due to the influence of the WLY feature. We expand the existing work on online dating by asking:

1. What is the impact of the *who likes you* feature on engagement, number of matches and match efficiency for online dating platform users?
2. What are the underlying mechanisms that drive the main effects of this feature?
3. What is the moderating role of the focal user’s and the targeting users’ desirability on the impact of the *who likes you* feature on user engagement and matches?

Based on prior literature in this domain, and established structural differences across genders in dating markets, we conduct our analysis separately for men and women (Fisman et al. 2006, Bapna et al. 2016). For the purpose of this study, we partner with a large North American online dating platform that we call *monCherie.com*, to run a randomized field experiment. In this experiment, we follow 100,000 newly registered users, and gift a random sample of 50,000 users the *who likes you* feature. We note that our key outcome variables are count variables and therefore we used a negative binomial model to quantify our treatment effects. Our findings show that the

treatment causes women to be more proactive, sending 7.4% more messages. This is a market improvement given that men send double the number of messages as compared to women, an asymmetry often attributed historical social norms around gender. Further, we find that the women endowed with this feature increase their matches by 14.4%, while men increase their matches by 11.5%. An interesting aspect of our findings is that the underlying mechanisms causing an increase in matches are not purely based on increasing engagement, which is often a dominant force in the narrative around digital communities. For instance, if engagement was the sole mechanism at work we would expect to see an increase in matches initiated by the focal user. While this happens under treatment in our setting, we also, interestingly, see an increase in matches initiated towards the focal user. We unpack this via more nuanced signaling story, unique to online dating, using two additional micro-level constructs detailed in Section 5.2.

One of the core features of WLY is that focal users can gauge the desirability of the users who send the “like,” as consequence of the revelation of the targeting user’s identity. Our heterogeneity analysis models the three-way interactions between the treatment, the focal user’s desirability and targeting users’ desirability. We find evidence that this feature drives sorting, encouragement, and discouragement depending on the combinations of the levels of desirability of these two sets of users. We see a positive and significant increase in matches for women when a woman focal user and the male targeting users have comparable desirability. This is evidence that the WLY feature is associated with an increasing level of sorting with respect to desirability levels. In the case where there is a disparity in desirability—when the targeting user has high desirability and the focal woman has medium-desirability—we see an increase in matches (for the focal user), providing evidence of an encouragement effect. Finally, in the extreme case of disparity in desirability, when the targeting users have high desirability and the focal woman has

low desirability, this effect becomes negative. This is evidence of a transition from encouragement effect towards market sorting on the part of the focal user, where users seem to be cautious about overreaching. Finally, when there is disparity because the targeting users have low desirability but the focal women are either medium- or high-desirability, we again see a negative effect on matches. We observe a discouragement for women who are desirable—i.e., they initiate fewer matches—when they are being voted on primarily by less desirable men. Our main finding is that the WLY feature has nuanced mechanisms through which it plays out, conditional on the desirability of the parties involved.

We extend the existing work on identity disclosure in online markets, which has looked at, for instance, the impact of reviewer identity disclosure on product sales (Forman, Ghose, Wiesenfeld 2008) and the negative influence of hiding identity information on subsequent crowdfunding donations (Burtch, Ghose, and Wattal 2016). This is in the context of users making a purchase or spending decision that are governed by either utilitarian or pro-social norms, which cannot be generalized to online dating matching markets. That is, how one behaves on Amazon or Kickstarter does not reflect how one will behave on Tinder or Match.com. To the best of our knowledge, little is known about how this might play out in matching markets where hedonic aspects of human interaction play a role. Altogether, our study contributes to the two-sided online platforms literature, where we meticulously show the ramifications of reducing information asymmetry between both sides of users under the influence of our WLY treatment.

The remainder of the paper is organized as follows. Section 2 provides a brief review of the literature. Section 3 describes the institutional details of monCherie.com, the online dating platform that we partner with to conduct our randomized experiment. Section 4 explains the setup of the randomized experiment as well as some empirical regularities. Section 5 describes our

results and robustness checks. Finally, in Section 6 we conclude and provide a discussion of our findings.

2. Background Literature

Our study is at the intersection of literatures related to dating markets and signaling in online markets. An important aspect of these markets is the heterogeneity in behavior across different individuals based on their characteristics, such as gender and desirability. The literature on dating and marriage markets inform us about gender differences prevalent in those markets due to asymmetric preferences and different actionable choices of both genders. Prior research has demonstrated that women and men have asymmetric preferences for characteristics of a potential partner: women place a higher importance on intelligence (Fisman et al. 2006), income and education (Hitsch et al. 2010), while men value appearances (Fisman et al. 2006). Besides preference differences, the offline dating and marriage markets foster an array of social frictions not only due to high search costs, as mentioned before, but also due to prevalent social norms. While online dating platforms can reduce search frictions, the social frictions stemming from gender differences in preferences and social norms still exist. For instance, the social norm that inhibits women from making the first move in dating is still evident in online dating markets (Piskorski 2012, Bapna et al. 2016).

Further, in dating markets desirability of participants is valuable. In online dating, the profile picture as well as entire profile page with personal information can convey desirability of the participants. The dating literature reveals that individual attractiveness, a key driver of individual desirability, matters when seeking romantic-partners albeit often asymmetrically across genders (Eastwick and Finkel 2008, Reis et al. 2013). Bruch and Newman (2018) find that dating platform users pursue partners who are more desirable than themselves, but chances of receiving

a response falls severely as the pursued mate is increasingly more desirable than the pursuer. This result suggests that desirable individuals are both pursued in greater numbers by others, and that they are more prone to rejecting the offers from the pursuers. Given our manipulation's immediate effect is likely to anchor the focal user on the profile picture of the targeting user and its desirability, we are motivated to examine the heterogeneity in treatment effects as a function of the desirability of both sides.

Evidence from extant literature point towards the high demand for attractive individuals. On the one hand, attractive individuals fare better in life outcomes than unattractive individuals (Eagly et al. 1991, Langlois et al. 2000). Moreover, research in social psychology has linked attractiveness with boosting confidence in romantic settings (Murphy et al. 2015, McGloin and Denes 2018). Therefore, in the online dating environment, we expect that likes from desirable targeting users can increase the focal user's self-esteem. This leads to an "encouragement effect," whereby the focal users feel encouraged to increase their activity levels on the dating platform, reaching out to desirable mates. Conversely, the focal user getting a like from a less desirable individual is likely to lead to a discouragement effect.

In summary, we contribute by filling extant gaps in the literature around the understanding of the efficacy of new digital signaling mechanisms in the context of one of most fundamental human activities. The primary, and first-order effect of this signal is likely to be hedonic in nature, focusing on individuals' desirability. Thus, we examine heterogeneity in treatment effects as a function of focal user's and targeting users' desirability. To the best of our knowledge, these are uncharted territory for the extant literature.

3. Institutional Details

To examine the relationship between the ‘*who liked you*’ (WLY) feature and engagement, number of matches, match efficiency, and match sorting outcomes, we conduct a field experiment in partnership with one of the largest online dating platforms in North America, which we call *monCherie.com* (name disguised). This site offers the following features to its users, which are typical of most other online dating platforms:

- Users set up their own well-structured online profiles where they describe themselves as well as reveal characteristics sought in a desired partner. Users may also include a set of their photos in their viewable profiles.
- Users can view profiles of all other users without limitations.
- Users search for profiles of other users using an advanced search engine that allows filtering by age, location, religion, and a number of other demographic variables. Users may also discover partners using the platform’s proprietary recommendation engine.
- Users can send private messages to any other user.
- Users can send a ‘like’ to other users when they visit their profile pages. In the online dating platform’s proprietary recommendation engine that allows users to discover partners by showing them profiles of potential matches, users can “like” or “pass” when they are presented with potential matches. Further, in the mobile phone version of the platform, users can swipe right in order to “like” other users, and swipe left to “pass.”

4. Data and Methodology

4.1 Data & Randomization

The experiment is conducted on 100,000 new users of *monCherie.com* over three consecutive months, referred to as month 1 (pre-treatment), month 2 (during treatment) and month

3 (post-treatment). A randomly selected treatment group of 50,000 new users from the initial pool of users are treated with a free gift of the WLY feature for one month (month 2), while the remaining users stay with the default settings (i.e. without access to the WLY feature) and serve as our control group. Note that monCherie.com constitutes a typical freemium community: most of the users sign up for a free account, which allows them to utilize all the key features listed above. Users can also obtain a premium subscription for a monthly fee, which consists of a fixed bundle of premium features that include, among other incremental features, the WLY feature, which is the ability to view the identity of the targeting users who have “liked” the focal user. For this study, we only sample from the non-premium free users to avoid selection bias. We are also able to determine whether the users are valid (valid = 0 represents an invalid user, if the user is a spammer or a bot as determined by internal algorithms at monCherie.com), and limit our sample to users who are valid, active (defined as users sending at least one view in the pre-treatment period), and heterosexual. This set of filters yields a final total of 41,500 users for the purpose of analysis.

The reason for limiting based on sexuality is that there are structural differences between heterogeneous and LGBTQ+ dating markets (Conway et al. 2015, Gobrogge et al. 2007) with respect to preferences, theoretical mechanisms and resulting behaviors. For instance, let’s take the well-known age preference relationships established in the heterosexual dating markets literature. Evolutionary psychologists argue that men have a preference for younger women as they perceive age to be a cue for fertility with younger women having higher reproductive value (Buss and Schmitt 1993). Conversely, women prefer older men, because age might be a cue to male socioeconomic status and income. These theories have been well tested with data from speed dating (Fisman et al. 2006) and online dating (Hitsch et al. 2010), with heterosexual individuals.

In contrast, Conway et al. (2015) provide strong evidence that in homosexual markets both men and women have a much wider age range preference than in heterosexual markets. And age is only one of the many dimensions that make up the composite preference structures of dating and mating preferences. We do not have any foundational literature, or measurement in our data for other dimensions previously studied in the heterosexual literature (Fisman et al. 2006) such as intelligence, racial preferences, and ambition, nor for latent gender identity within homosexual markets. Thus, any findings that we present will be in a vacuum and would be a dis-service to the homosexual community. Finally, we also note that from a practical perspective, our sample size for homosexual population is close to 2200, as opposed to 41,500 for heterosexual users. Not surprisingly, when we run our analysis through this much smaller sample size we find virtually no significant treatment effects. It is unclear to us what meaningful conclusion we can draw from such a null result, and we leave this very important topic as the subject of future research.

The WLY treatment is randomly and simultaneously assigned on the sample of new and free users, allowing us to causally identify the treatment effects on the treated population. As a result, our study is free from endogeneity concerns, and other non-random differences that would otherwise need to be controlled for explaining any observed differences between the treatment and control groups. Furthermore, the treatment is endowed on users by the dating platform. That is, the treated users receive the treatment as a gift, without any required actions on their side, and are not asked for anything in exchange. Therefore, users are unaware of being a part of an experiment, so observer bias is not applicable.

In addition to the data from the 100,000 random users who are part of the experiment, we also collect data about the users who directly interact with the focal users in this experiment. For each of the 100,000 users in our experiment, we have time-stamped viewing and messaging data.

Further, critically for this study, we also have time-stamped likes (i.e. “liking” is 1, “not liking” is 0, where “not liking” is not visible to users of the platform), both sent and received by these users. Further, our dataset consists of data from the 5,640,774 additional users who interact with the focal users through the engagement channels, like viewing, messaging, or liking.

In addition, we have data on a set of demographic variables such as gender (female or male), age⁵, sexual orientation (straight, gay, or bisexual), ethnicity (Asian, Black, Hispanic, Middle Eastern, Native American, Pacific Islander, White, mixed race, other, or unknown), body type (average, curvy, fit, full figured, jacked, little extra, overweight, thin, used up, won’t say, or unknown), and education level (high school, two-year college, university, post-graduate, space camp, or unknown). Out of these, age groups, ethnicities, body types and education levels are all control variables that increase the efficiency of our treatment effect estimates. Our measure of desirability, following Bruch and Newman (2018) and Shi and Viswanathan (2020), is the number of unique targeting users from whom the focal user has received a message in the pre-treatment period. This measure is well defined for every user and can take on the value of any non-negative integer. Our analysis is conducted following the approach used in the signaling literature, which discretizes the measure described above—unique targeting users from whom the focal user has received a message—into tertiles or three buckets labeled low, medium, and high (based on the empirical frequency distribution separately for both genders). Following Bruch and Newman (2018), our measure essentially proxies for the user’s overall desirability. The intrepid reader will see the difference between how desirability is perceived in the treatment (with pictures) and control (blurred pictures) group based on Figure 1 above.

⁵Based on the relationship science literature cited in the literature review section above, age is classified into seven age-groups below 18, 18 to 24, 25 to 29, 30 to 39, 40 to 49, 50 to 69, and 70 and above.

Tables 1 and 2 provides pre-treatment comparison of the treatment and control groups separately for both genders. We see that the treatment and control groups are statistically indistinguishable from each other both in individual characteristics and in online dating engagement metrics. The three key metrics summarized in Table 1 are *Views*, *Messages*, and *Matches* for the pre-treatment month. To operationalize a match, we rely on other work on online dating that defines a “match” as the exchange of at least four messages between two parties (Bapna et al. 2016, Shi and Viswanathan 2020).⁶ Using this definition, we examine the impact of WLY treatment on the number of views, messages, and matches that are both sent and received by the focal users during the treatment month. In this study, *MatchesSent* are those where the focal user initiates the exchange of the four messages, and *MatchesReceived* are those where the focal user is the recipient of the first message in the four-message exchange. Collectively, we term these variables as online dating engagement (i.e. *ViewsSent*, *ViewsReceived*, *MessagesSent*, and *MessagesReceived*), and matching (i.e. *MatchesSent* and *MatchesReceived*) variables (following the naming conventions used in Bapna et al. 2016). To further increase the precision of our treatment effect estimates, we use the first principal component of the pre-treatment online dating engagement and matching variables as another control variable.⁷ Turning to the individual characteristics reported in Table 2, we notice that there are stark differences between men and women with regards to desirability scores. Though we are using a different measure, recall that this is consistent with Rudder (2014) who shows that female users rate male users more harshly.

⁶ Prior research (Hitsch et al. 2010, Bapna et al. 2016) and anecdotal evidence from online dating industry has pointed out that exchange of three messages between potential couples is a good predictor of actual online match, when participants exchange phone numbers or ask their potential mate out for date during an online dating message exchange session. Accordingly, in this paper we conservatively choose exchange of four messages as definition of match.

⁷ The pre-treatment online dating engagement and matching covariates are highly correlated. Using them separately in our estimation model leads to multicollinearity issue. In place, using the first principal component, which explains 64% variance and 70.7% variance for the female and male samples, respectively, avoids this issue.

This validates our desirability measure, which is based on pre-treatment messaging behavior following the signaling literature (Bruch and Newman 2018).

Examining the baseline metrics of the control group in Table 3, we see that men send approximately two times the number of messages that women send (~12 messages vs. ~6 messages, respectively), and women receive on average, seven times the number of messages as compared to men (~22 messages vs. ~ 3 messages). This implies that multiple men are targeting the same smaller set of women. Overall, this shows us that men are much more active on these platforms and more generally, supports prior work that shows that men and women engage in these dating and marriage processes in different ways. Table 3 also includes preliminary t-tests that are usually used to analyzing online experimental data (Deng et al. 2013, Bapna et al. 2016). However, it is important to note that our key outcome variables are count variables, with unequal means and variances, and therefore we used a negative binomial model to quantify our treatment effects.

4.2 Methodology: Negative Binomial

The original theoretical regression model of interest is

$$Y = X\beta + \epsilon, \quad (1)$$

where Y is an $N \times 1$ column vector of the outcome of interest, X is a $N \times k$ matrix of covariates (including treatment, primarily, but also control variables and a column of ones for the intercept term), β is a $k \times 1$ column vector of regression coefficients, and ϵ is a $N \times 1$ column vector of random error. Although Y are counts, we could (and did) first estimate β in (1) with OLS; in obtaining this $\hat{\beta}_{OLS}$ it becomes clear that our data failed to conform to these modeling assumptions. Specifically, when Y is assumed to be distributed Gaussian (which is common given the relationship between OLS estimation and Gaussian noise, though not strictly required), then we also know that the errors (i.e., $\frac{(Y_i - \hat{Y}_i)^2}{Var[Y_i]}$ where $\hat{Y}_i = X_i \hat{\beta}$) are distributed χ^2_{n-k} . When we

compare our errors to this theoretical distribution we find extremely significant evidence (at a 0.05 significance level) that our errors are not distributed χ^2_{n-k} , meaning that the errors in (1) are unlikely to be Gaussian. We then estimate β in (1) with a generalized linear model (GLM) with a Poisson error distribution (i.e., family). We can evaluate the overall performance of this model with the Pearson Test statistic for the errors (i.e., $\sum_{i=1}^n \frac{(Y_i - \hat{Y}_i)^2}{\hat{Y}_i}$) or the deviance statistic. For clarity, the deviance is twice the maximum achievable log-likelihood and the log-likelihood of the fitted model; therefore, it will take a specific form depending on the family of the chosen GLM. Both statistics are distributed χ^2_{n-k} , and using both tests, we find statistically significant evidence (at a 0.05 significance level) of a lack of fit for the Poisson model. An implicit assumption of the Poisson distribution is that the mean and variance are equal, and violations of this assumption (i.e., dispersion) can lead to the poor fit we observe. Therefore we instead consider estimating β in (1) using a negative binomial error distribution. The negative binomial does not require this relationship between the mean and variance, as it has an additional parameter, which can be used to adjust the variance independently of the mean. We compute a likelihood ratio test of the Poisson and the negative binomial models, under the null hypothesis that there is no difference between the two, and find significant evidence (at a 0.05 significance level) that the negative binomial is a better fit for the data. Additionally, the deviance goodness-of-fit for the negative binomial does not find significant evidence (at a 0.05 significance level) of poor fit. Therefore, going forward, we elect to represent estimate β in (1) by $\hat{\beta}_{NB}$ from a GLM with the negative binomial family.

We estimate the negative binomial models separately for men and women following prior literature (Bapna et al. 2016). We confirm the appropriateness of this approach by comparing the AIC of the joint model with men and women pooled to the joint AIC of the separate models.

We find that the joint AIC of the separate models indicate that separate models are a better fit for the observed data, once we have accounted for the additional parameters used in estimating the separate equations.

Another issue in dealing with experimental data from two-sided markets is the possibility of treatment interference due to competition amongst users for matches. Recent research has shown that average treatment effect estimates in online marketplaces can be biased due to Stable Unit Treatment Value Assumption (SUTVA) violations, which primarily result from interference (see Blake and Coey 2014, Fradkin 2015, Holtz and Aral 2020). For instance, consider a car sharing platform, say Uber, running an A/B test (i.e., a randomized controlled trial) to examine a price discount. An increase in demand in the treatment group due to the price discount could spill-over into the experience (say, longer wait times) of the control group users. This would happen due to the presence of a shared resource, the constrained pool of drivers in a given geography, between treatment and control groups. Such a spillover would clearly over-estimate the treatment effect of the price discount. It is theoretically possible that there exists interference between treatment and control users in the dating context, which could bias our treatment effect estimates if not addressed. However, in contrast to the Uber example above, SUTVA is not a concern in our context. Firstly, given the overall size of the market of our research site, the 100,000 users in our experiment are a tiny fraction of the overall set of users on the platform (~1.8 million)⁸. Secondly, the mean percentage of treated neighbors of our experiment users is 1% and 0.55% for women and men respectively, and the median is 0 for both (Table B1 in the Appendix B).⁹ This is indicative that our treatment is extremely sparse and therefore SUTVA violations are not a significant

⁸ <https://www.statista.com/statistics/826778/most-popular-dating-apps-by-audience-size-usa/>

⁹ To get this number, we construct a ViewsSentPre network using the pre-treatment views sent activity, where the target user to whom the focal user sent a view is considered a neighbor of the focal user. We do this separately for women (n=11,171) and men (n=28,549).

concern in this environment. Thirdly, the product, i.e. the users, are non-rival in that unlike Uber and Airbnb, where one person's usage of the ride (Uber) or the home (Airbnb) impacts another person's supply, this is not the case on online dating. In online dating, it is the norm that any given user will match and date multiple people at the same time.¹⁰

However, for robustness and for the benefit of future work where the context may be different, we implement a model with the SUTVA correction term (Chin 2019). This approach is detailed in Section 5.5.4 and Appendix B.

4.3 Methodology: For Heterogeneity Analysis

For our heterogeneous treatment effects analysis, we consider the moderating impact of the focal user's and targeting users' desirability on the WLY treatment effect. We calculate targeting users' desirability based on the average desirability of all users who target the focal user, and bin the averages into tertiles as described earlier.

To understand the interplay between focal user's desirability and targeting users' desirability, we run our negative binomial model adding the following terms: (1) indicator variables for the focal user's desirability (for both medium and high level of desirability); (2) indicator variables for the targeting users' desirability (for both medium and high level of desirability); (3) interaction terms for the interaction between the treatment dummy and focal user's desirability (for both medium and high level indicator variables); (4) interaction terms for the interaction between the treatment dummy and targeting users' desirability (for both medium and high level indicator variables); and, most importantly, (5) a set of three-way interaction terms that include the treatment dummy, the focal user's desirability, and targeting users' desirability (for both medium and high level indicator variables for each type of desirability). As a reminder,

¹⁰ <https://globalnews.ca/news/3943295/dating-more-than-one-person/amp/>

our measure of desirability, which follows existing signaling literature, is based on the level of attention the user received measured by the number of unique targeting users from whom the focal user has received a message in the pre-treatment period (Bruch and Newman 2018, Shi and Viswanathan 2020). We discretize the measure into tertiles labeled low, medium, and high (based on the empirical frequency distribution separately for both genders), and utilize dummy variables for medium and high tertiles in our model. As a construct, the low tertile of desirability is the baseline, and implied when both medium and high tertiles equal zero.

Following the online dating literature (Bapna et al. 2016, Shi and Viswanathan 2019) and our main analysis, we also use separate gender models for our heterogeneity in treatment effects analyses centered on desirability. We note that for this heterogeneity analysis, there is a tradeoff that arises between modeling the genders in a single model (with a gender indicator variable) or estimating models on each gender's data separately. As in our main analysis, a single (joint) model there is a potential for biased point estimates and biased variance-covariance estimates, because such a model would impose strict assumptions about the homogeneity (between genders) of critical model parameters (e.g., between focal user's and targeting users' desirability). Utilizing separate models should ameliorate the concern of biased point estimates, but it may not alleviate concerns about the biased variance-covariance structure. Moreover, it can exacerbate them because in our separate models, we are marginalizing over the inherent relationship between the genders, i.e., the covariances of the parameters related to the other gender's desirability. Note that this is not relevant in our base models where the treatment effect estimate jointly marginalizes over both genders' desirability parameters. However, in the heterogeneity analyses, the separate gender model approach has to be examined with respect to its impact on the significance of the point estimates. As mentioned earlier, we compare between the AIC (Akaike information criterion) for the pooled

gender model and the combined AIC for the separate gender models.¹¹ The combined AIC for the separate gender models is less than that of the pooled model (See Table A1 in the Appendix A), which indicates that separate models are a better fit for the observed data, even after penalizing the additional complexity (parameters) when estimating the separate equations. Therefore, the AIC indicates that the point estimate benefit we obtain by separating the genders — the removal of finite sample bias and guarantee of consistency—exceeds the potential inaccuracies we incur in the covariances we used in the hypothesis tests, especially at large samples size, as in our setting.

5. Results

5.1 Main Treatment Effects

We first consider the main treatment effect results for females as shown in Table 4. Based on our negative binomial model, we find that there is significant uptake in self-initiated engagement outcomes for females due to the WLY treatment. Specifically, *ViewsSent* significantly increased by 10.4% (Table 4, column 2; $p < 0.05$) and *MessagesSent* significantly increased by 7.37% (Table 4, column 4; $p < 0.1$).¹² When these percentages are applied to the baseline average outcomes in the control group (Table 3), we see that there are 2.8 additional *ViewsSent* (where the baseline is 27 *ViewsSent*) and 0.42 additional *MessagesSent* (where the baseline is 5.7 *MessagesSent*) for females due to the treatment. Accordingly, we witness that our treatment causes women to become proactive, as they are sending more views and messages. This is a highly desirable improvement for dating markets, as prior literature has established that self-initiated engagement outcomes are rare for females in two-sided dating platforms.

¹¹ For the separate gender models, we construct the combined AIC by adding the log-likelihoods across the two models, as well as we add the penalty terms (since the number of parameters are also additive).

¹² Calculation of the percentage value uses the following formula: $|e^{\beta_{coef}} - 1| \times 100$

Besides an increase in self-initiated engagement outcomes, the treated females also witness an increase in engagement received from other users due to the WLY treatment. We see a significant increase by 5.2% for *ViewsReceived* (Table 4, column 1; $p < 0.05$) and by 5.6% for *MessagesReceived* (Table 4, column 3; $p < 0.05$). Turning to matching outcomes, we find that, for the average female user, an increase in matches is derived from both an increase in *MatchesReceived* and *MatchesSent*. That is, treated females see a 14.28% increase in *MatchesReceived* (Table 4, column 5; $p < 0.01$), and a 14.41% increase in *MatchesSent* (Table 4, column 6; $p < 0.05$). This translates into an additional 0.3 *MatchesReceived* as compared to the baseline average of 2.3 in the control group, and an additional 0.06 *MatchesSent* against the baseline average of 0.4 in the control group.

Table 5 reports the main treatment results for males. Looking at engagement outcomes, we observe a significant increase in views and messages (both sent and received) due to the WLY treatment. Interestingly, we find that *MessagesSent* increased by 8.93% (Table 5, column 4, $p < 0.01$) for treatment males, while *MessagesReceived* significantly increased by 7.25% (Table 5, column 3, $p < 0.01$) for treated males. This shows that, on average, treated males send 1.1 additional messages against their baseline average of 12.1 in the control group, and they receive 0.22 additional messages as compared to their baseline average of 3.1 in the control group. As a result, overall engagement for both genders increase significantly due to the WLY treatment.

Besides an increase in the engagement outcomes, treated males show a significant increase of 14.59% in *MatchesReceived* due the treatment (Table 5, column 5; $p < 0.01$). They also show a significant increase of 11.51% in *MatchesSent* (Table 5, column 6; $p < 0.01$). Even though these are substantial increases in percentage terms, in absolute terms we see an increase in *MatchesReceived* to 0.22 from a baseline of 0.19 (i.e. an additional 0.03 units). Similarly,

MatchesSent increases to 1.3 from a baseline of 1.2 in the control group (i.e. an additional 0.1 units). All of the results, with respect to females and males, are over the course of just the users' second month on the platform; if the WLY treatment was experienced earlier and/or longer the effects could be larger.

An interesting puzzle in our results is that for both genders, our treatment significantly impacts *MatchesReceived*. If engagement was the sole mechanism of the treatment effect, we would expect to see an increase in *MatchesSent*, based on extra engagement of the focal user who receives the treatment. We do observe an increase in *MatchesSent*, however, the fact that the treatment effect is significantly discernable also for *MatchesReceived*, points to a subtler signaling story, which we investigate and decompose into in the next section.

5.2 Additional Signaling Mechanism

The underlying mechanism driving the effect on *MatchesReceived* is similar for men and women. To unpack the underlying mechanism, we create two new constructs, *VoteViewsSent* and *VoteViewMessagesReceived*.

- *VoteViewsSent* is the number of unique other users that the focal user sent a view to during the treatment period, after they received their first like from a targeting user.
- *VoteViewMessagesReceived* is the number of unique messages the focal user receives as a result of *VoteViewsSent*. More specifically, it is the number of unique other users that message the focal user, after the focal user sent them a view, specifically after the focal user received his or her first like from a targeting user. This count excludes any other users who sent messages to, or received messages from, the focal user prior to the sent vote view.

The goal of these two new constructs is to explain why the treatment (the focal user being able to see who liked them) could lead to an increase in *MatchesReceived*. A focal user, observing that

identity of someone likes them, may see this “like” as a signal that can create an encouragement effect. This encouragement results in the focal user subsequently engaging with other users they may be interested in, e.g. through sending views. These views sent by the focal user, to the other users, triggers the other users to send a message to the focal user. This begins the matching process, which if successful, ends with matches received by the focal user.

Summary statistics for these advanced outcome variables are provided in Table 6. Interestingly, the odds of receiving a message in response to a sent vote view is 18.7% for female focal users but 4.5% for male focal users, which reflects the inherent asymmetry present in dating markets along gender lines.¹³ The results of our estimation models are reported in Tables 7 and 8. For women (Table 7), we find that the increase in *MatchesReceived* is in essence driven by the sequence of actions captured in the two new constructs described above. That is, we see a significant increase in both *VoteViewsSent* and *VoteViewMessagesReceived* as a result of the treatment. We observe that a treated female focal user sends 10.32% ($p < 0.05$) more views (*VoteViewsSent*) in response to a “like” and receives 11.57% ($p < 0.01$) more messages (*VoteViewMessagesReceived*) in return from those views as compared to the control group. This translates to 2.8 additional *VoteViewsSent* and 0.6 additional *VoteViewMessagesReceived* against the baseline of 26.8 and 5 in the control group, respectively. This provides one underlying mechanism behind the observed increase in *MatchesReceived* for women.¹⁴ Similarly, for men (Table 8) we see a 7.09% ($p < 0.05$) and 8.97% ($p < 0.01$) increase in *VoteViewsSent* and

¹³ Percentage calculation uses $\frac{VoteViewMessagesReceived}{VoteViewsSent} \times 100$.

¹⁴ We acknowledge that other mechanisms can also explain the effect on *MatchesReceived*. Future work will need to separately design interventions to isolate various differential mechanisms at work.

VoteViewMessagesReceived, respectively. This provides support for these two mechanisms also underlying the increase in *MatchesReceived* for men as well.¹⁵

To summarize, we observe an indirect effect of the WLY treatment on both genders *MatchesReceived* through the sequence of actions described above. Next, we look into the match efficiency and time to first match related outcomes that provide a more nuanced view on the impact of the WLY treatment.

5.3 Match Efficiency Effects

Beyond the main treatment effect analyses, to get a more nuanced view on online dating engagement and matching outcomes, we study the impact of WLY on match efficiency and time to first match. We construct the match efficiency outcomes as the ratio of matching to messaging separately for sent and received scenarios (i.e., *MatchesReceived* divided by *MessagesReceived* and *MatchesSent* divided by *MessagesSent*). In particular, we find that match efficiency significantly increases by 0.76% for females in the received scenario (Table 9, column 1, $p < 0.01$), but not for males (Table 10).

Next, we calculate ‘time to first match sent’ and ‘time to first match received’ as the time taken since exposure to the WLY treatment to get the first match sent and first match received respectively. Using a Cox proportional hazards model for estimation, we find that the time to first match received is significantly shorter for females by 11.69% (Table 9, column 3, $p < 0.05$). The positive and significant coefficient here implies that treated females take less time, on average, than the control group, to get their first match received, during the treatment period. For treated males, we find both the time to first match received is significantly shorter by 14.43% (Table 10,

¹⁵ For this study, we are less interested in clear identification in an econometric sense of these intermediate micro-level sub-constructs. Rather, our aim is to put forward a mechanism in play that can explain the effect on *MatchesReceived* due to the treatment.

column 3, $p < 0.01$) and time to first match sent is also significantly less by 9.86% (Table 10, column 4, $p < 0.01$). In summary, we find that the treatment causes women and men to get their first match relatively faster, which is a desirable outcome of the WLY feature. Finally, a key aspect of the WLY feature is that it reveals the desirability of the targeting users who gives likes, which we explore in the next sub-section.

5.4 Role of Desirability

The WLY feature, on the surface, is designed to reduce the information asymmetry in the matching process. Now, under the treatment of WLY, instead of a focal user knowing only the number of people who “liked” them, they can now also see the identity of those users, including the all-important profile picture and profile page, which increases the salience of the targeting users who send “likes.” But is this additional information necessarily uniformly good for everyone? To unpack this, we dive into the heterogeneity of the treatment effect, which factors in not only the focal user’s desirability but also the desirability of the targeting users who like the focal user.

The results from the heterogeneity analyses are in Tables 11 (female) and 12 (male). We find evidence of moderating impacts of desirability (a key aspect of our WLY feature): market sorting, an encouragement effect, and a discouragement effect depending on various combinations of the focal and targeting users' relative levels of desirability. For females, we see a positive and significant increase in *MatchesSent*, when female focal users and targeting male users have similar desirability levels – i.e. low-low, medium-medium, or high-high values (Table 11, column 2, rows 1, 14 and 17, respectively). Interestingly, we find a positive and significant increase in *MatchesReceived* for low-low and medium-medium desirability tertile pairs (Table 11, column 1, rows 1 and 17, respectively). These results show that our WLY feature leads to sorting, as treated

females initiate significantly more matches with potential matches of comparable desirability levels.

Next, we examine the results for scenarios where there is disparity in desirability between the focal and targeting user groups. We first consider cases where the targeting users' desirability is higher than the focal user's desirability. When the targeting male user has high desirability and the focal female has medium desirability, we find a significant increase in *MatchesSent* for the focal female user (Table 11, column 2, row 15). This provides evidence of an encouragement effect. In contrast, female focal users with low desirability do not show any significant effect with respect to their *MatchesSent* outcome when receiving likes from medium desirable targeting users, which suggests that the encouragement effect is applicable to female users above a certain threshold level of desirability (i.e. it exists for medium tertile female users only). In the extreme case, when female users with low desirability receive likes from highly desirable targeting users, these females significantly reduce their *MatchesSent* due to the treatment (Table 11, column 2, row 12). This suggests a transition from encouragement effect towards market sorting on the part of the focal female users, where those with low desirability seem to be cautious about overreaching. Further evidence of this transition is the negative and significant impact on the *MatchesReceived* outcome, when focal female users have low desirability and targeting male users have medium desirability (Table 11, column 1, row 13).

Finally, we look at the scenario when targeting male users are less desirable than the focal female users. For cases where the focal female users belong to high or medium desirability levels but targeting male users belong to low desirability level, we see a significant negative effect on *MatchesSent* for these female users (Table 11, column 2, rows 10 and 11, respectively). In addition, for focal female users of medium desirability and targeting male users of low desirability, we see

a significant negative impact on *MatchesReceived* (Table 11, column 1, rows 11). These results show a discouragement effect for treated females who are desirable above a certain threshold, but are being liked by primarily by males in the lowest tertile of desirability. Yet, when female focal users from the high tertile of desirability receive likes from male targeting users of medium tertile of desirability, we do not observe a significant effect on any matching outcomes (Table 11, row 16). These results convey that the discouragement effect appears to exist only when male targeting users belong to the bottom tertile of desirability.

For male focal users, for most combinations of focal user and targeting user desirability levels, we do not find any discernable moderation effects in play. One particular situation of interest shows up where we find market sorting. We find that there is a significant increase in *MatchesReceived* for male focal users in the lowest tertile of desirability when receiving likes from female targeting users who also belong to the lowest tertile (Table 12, column 1, row 1). In addition, we find support for discouragement effect in one situation, where under the treatment, focal male users of medium desirability witness a significant negative impact on *MatchesReceived* (Table 12, column 1, rows 11) when targeting female users belong to the lowest desirability tertile.

In summary, our results related to desirability show market sorting, encouragement, and discouragement primarily for female focal users, depending on the combinations of focal and targeting user desirability levels. To the best of our knowledge prior literature has not addressed this important heterogeneity based on desirability, a user characteristic that is central in dating markets.

5.5. Robustness

In this section, we perform a series of robustness checks to tackle various alternative scenarios that can explain the treatment effect. At first, we rule out alternate explanations for the treatment effect.

Here, we first show that there is no treatment effect in the post-treatment month since the platform users restore to their usual behavior once the treatment expires, ruling out alternative explanations for our results. We then ask whether granularity in the timeframe leads to temporal variations in treatment effects, and conduct analyses at the weekly level. Additionally, we craft a set of two robustness tests to check whether the treatment effect is a result of dating platform users reacting to the WLY treatment as a free gift or due to the treatment increasing salience of the feature. Finally, we present results of a SUTVA corrected model.

5.5.1 Ruling out Alternative Explanations

As we find in this study, the significant difference between treatment and control groups only exists due to the WLY manipulation. In contrast, if there exists an alternative explanation for the exhibited treatment effects, we would expect that the treatment effect will continue even during the post-treatment period. Therefore, to test for the presence of another explanation, we re-run the negative binomial model on the post-treatment month dataset for both females and males separately. The results show that there is no significant impact of the WLY manipulation on the *MatchesReceived* outcome variable for both females (Table 13, column 2) and males (Table 14, column 2). This confirms that the difference between treatment and control groups that we observe during the treatment period is solely due to the effect of the WLY manipulation, and not because of an alternate explanation.

5.5.2 Examining Weekly Level of Analysis

We conduct our study at the monthly-level as the unit of analysis. This does not allow us to check for temporal variations in treatment effect at a more granular timeframe. It may be possible that at the weekly level, the treatment effect from the WLY manipulation may be inconsistent with our findings at the monthly level. Therefore, as a robustness check, we calculate the treatment effect

at the weekly level of analysis using the negative binomial model, and report our results in Table 13 and 14. In Table 13, the results show that for the female population there is positive and significant increase in *MatchesReceived* for most of the weeks (i.e. week 1, 2 and 4) for the treatment month (Table 13, columns 3, 4 and 6). Specifically, there is a 18.06% increase (at $p < 0.01$) in the outcome variable in Week 1, and 14.69% increase (at $p < 0.05$) for Week 4, showing a constant trend in the treatment effect, which is comparable to the 14.28% increase that we find at the monthly level of analysis (Table 13, column 1). For men, we find a significant increase in *MatchesReceived* variable for weeks 1, 2, and 3 of the treatment month (Table 14). Lack of significance in the outcome variable for week 4 of month 2 is possible due to diminishing time trend in users' activities, as dating website activities in general reduce over time, as users stay in the platform extends. In summary, our weekly-level analyses and monthly-level analyses concur in their findings, showing similar pattern in the treatment effect across different unit of analyses.

5.5.3 No Gift Effect

Online dating platform users may be influenced by the WLY manipulation due to the “gift effect,” rather than the actual salience of the treated feature. As a consequence, treated users of the dating platform could react to receiving the gift only, resulting in an impact on engagement and matching outcomes. If there is a gift effect, we would expect that the treated users' activity metrics would go back to the usual levels when the novelty effect of the gift recedes. To check whether the treatment effect is driven due to the WLY manipulation being a gift or because of its own salience, we conduct two robustness checks. First, we compare treatment effects between week 1 and week 2 of the treatment month. If the treatment effect persists in week 2 of month 2, we can infer that the effect is due to the treatment itself, and not induced from it being a gift. We would expect that the novelty of the gift would recede soon (e.g, by one week), given that online platforms usually

indulge their users with a plethora of IT features (Kohavi and Thomke 2017). Second, as an additional check, we also compare the treatment effect between the last week of the treatment period (i.e. month 2, week 4) and the first week of the post-treatment period (i.e. month 3, week 1). In this scenario, we expect the gift effect will not suddenly disappear at the beginning of the post treatment period, as the essence of receiving a gift can continue even when the actual feature no longer exists. The results for the first robustness test rules out the gift effect since the treatment effect persists in week 2 of month 2 for both female and male samples (Table 13 column 4 and Table 14 column 4). For the second robustness test, we find that the treatment effect vanishes abruptly in the first week of the post-treatment month, while being present in the last week of the treatment month for the female sample (Table 13 column 6 and 7). For men, this analysis cannot be done as they show no treatment effect for the last week of the treatment month (Table 14 column 6), due to diminishing time trend in their online dating activities. In summary, our robustness analyses conclusively help rule out the gift effect explanation.

5.5.4 SUTVA Correction

As mentioned earlier, the online dating market fundamentally deals with a “good” that is non-rival. That is, one user’s consumption of a match does not affect the supply for others. In the United States, the context of our research, industry reports indicate that dating multiple users is the norm¹⁶. For sake of completeness and for future research that deals with other context where this may not be the norm, we also present SUTVA- corrected results based on Chin (2019).

We calculate a correction term based on the underlying directed networks for each of our outcome variables, and add it as a control variable in NB models following Chin (2019). For example, in the matches sent network, the correction term is the proportion of the focal user’s

¹⁶ <https://globalnews.ca/news/3943295/dating-more-than-one-person/amp/>

neighbors who have received matches from at least another user from a target cohort. Note that the target cohort of users depend on the focal user’s cohort. For example, if the focal user belongs to the treated female cohort, then the target users belong to the control female cohort. In other words, at least one individual from the target cohort and the focal user have both sent matches to a common neighbor. We term such a neighbor as an infected neighbor. Therefore, the correction term is the total number of infected neighbors divided by the total number of neighbors for a focal user. Further, the correction term calculation here is the proportion of the treated female focal user’s neighbors who have received matches from at least one control group female user, in the pre-treatment period. Our main results are robust to the SUTVA correction, as detailed in Appendix B.

6. Discussion and Conclusion

The process of selecting a partner is often complicated by the information asymmetries between individuals and the known gender asymmetries in preferences and social inhibitions prevalent in dating markets. With the emergence and growth of online dating platforms, we see these platforms implement technology-enabled features that help reduce these information asymmetries. In this paper, we study an IT-enabled feature that reveals ‘*who likes you*’ (WLY) in an online dating market, and look for its impact on an array of online dating outcomes. Utilizing an in-vivo randomized field experiment, we causally examine the treatment effects on engagement outcomes, matching outcomes, and match efficiency outcomes. Given that desirability of participants is central to dating decisions, a key aspect of our work focuses on heterogeneity of our manipulation as a function of the two users’ desirability.

To quantify the treatment effect of the WLY feature, we use a negative binomial model, as our outcome variables are overdispersed count variables. Overall, we find that female focal users

treated with this feature achieve 14.28% more *MatchesReceived* and 14.51% more *MatchesSent* than the control group, while male treated users experience a comparable increase in *MatchesReceived* to the order of 14.59% and *MatchesSent* by 11.51% in comparison to the control group. For females, we find an increase in self-initiated views and messages. This is particularly interesting given that women are typically known not to initiate communication in online dating situations as compared to men.

For the average user, the impact of the WLY feature plays out via both an increase in *MatchesReceived* as well as *MatchesSent* for both genders. The increase in *MatchesSent* is explainable with the known engagement-based explanation of user behavior in online platforms, yet increase in *MatchesReceived* is counter to the conventional intuition, thus requires further explanation. To unpack this, we explore an underlying mechanism driving this matching outcome using two new constructs—*VoteViewsSent* and *VoteViewMessagesReceived*. We observe that the increase in *MatchesReceived* for treated users across both genders is in essence explainable due to a direct increase in views sent in response to receiving a like, and messages received in response to those views. More specifically, the increase in *VoteViewsMessagesReceived* is a result of the *VoteViewsSent* triggering the weak-signaling mechanism, i.e. the digital signal sent with a profile view, as documented in Bapna et al. (2016).

Importantly, the WLY feature allows focal users to see the profile pictures and profile pages of those who send a like. Looking only at the profile pictures can reveal attractiveness information (McGloin and Denes 2018), yet additional access to entire user profile page reveal desirability of the targeting users that we further use to look for heterogeneity in our feature's treatment effect. We study the moderating effects of two type of desirability—the focal user's desirability and the targeting users' desirability, and find evidence for sorting, encouragement

effect, and discouragement effect for different combination of user desirability tertiles. We predominantly find heterogeneous results for female focal users. Interestingly, we find evidence of sorting for comparable desirability across all levels— low-low, medium-medium, and high-high. When there is disparity in desirability, we observe different effects depending on the direction of the disparity ranging from encouragement, to a sorting barrier, to discouragement. Future research, we hope, will be able to unpack why we observe these specific types of heterogeneity. That said, from a practical perspective, our finding provides a note of caution for platform management and policy agencies as implementation of IT-enabled features can have divergent and sometimes detrimental impact on specific cohorts of users.

Our results reveal that in these complex human related matters, how these features will impact behavior is not straightforward. Our paper sorts out various mechanisms at work, highlighting the important role of desirability, particularly for females, in matching outcomes on online dating sites. While existing work on online platforms more generally points to the importance of engagement, we show in the matching context the engagement story is not the full story. The importance of desirability is not only central in an online dating context, but also other matching markets where desirability of a potential job or employer, or a potential client will play a role in the matching prospects. If users pay for a premium feature such as WLY on LinkedIn, and they see only recommendations that fall in the lowest tertile of “desirability,” they may have similar responses. Future research can examine these questions in other matching contexts.

That said, we continue to be cautious in extrapolating from dating markets that have a very specific hedonic, primal and evolutionary psychology based grounding to more practical but structurally similar markets, say such as LinkedIn or employment matching portals. No one will argue that peoples’ behaviors in these markets are going to have a lot in common. Of course, the

analytical part of our experimental design and our analysis are general and should be useful to other researchers studying matching markets.

Broadly speaking, the results of this study demonstrate that technology-enabled features can have an important impact on matching markets and therefore have implications for the design of such platforms. Online dating platforms face the unique challenge of wanting to maintain users while still providing them a successful service, i.e. finding them a match. While these two objectives are seemingly in conflict with one another, our discussions with monCherie.com management indicate that the incentive to match people successfully is high as it creates offline word-of-mouth and generates a new set of users. Thus, implementing these technology-enabled features in such a way that can maximize the benefit of the users is important, and the results of this study can help online dating platforms do that.

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Figures and Tables

Table 1 – Randomization Check (Activity in Pre-Treatment Month)

Gender	Basic Outcomes	Manipulation	Observations	Mean	Std. Error	t-statistic	p-value
Females	<i>ViewsSent</i>	Treatment	5614	55.413	100.034	0.8313	0.4058
	<i>ViewsSent</i>	Control	5557	53.719	114.764		
	<i>ViewsReceived</i>	Treatment	5614	188.768	235.842	0.5514	0.5813
	<i>ViewsReceived</i>	Control	5557	186.276	241.830		
	<i>MessagesSent</i>	Treatment	5614	13.397	23.331	0.9504	0.3419
	<i>MessagesSent</i>	Control	5557	12.979	23.145		
	<i>MessagesReceived</i>	Treatment	5614	50.392	72.994	0.5088	0.6109
	<i>MessagesReceived</i>	Control	5557	49.686	73.782		
	<i>MatchesSent</i>	Treatment	5614	1.116	3.386	-0.1369	0.8911
	<i>MatchesSent</i>	Control	5557	1.125	3.641		
	<i>MatchesReceived</i>	Treatment	5614	6.454	12.039	1.0749	0.2824
	<i>MatchesReceived</i>	Control	5557	6.210	11.904		
Males	<i>ViewsSent</i>	Treatment	14291	88.332	187.968	-1.1637	0.2446
	<i>ViewsSent</i>	Control	14258	91.056	207.066		
	<i>ViewsReceived</i>	Treatment	14291	24.855	47.116	-2.2487	0.0245
	<i>ViewsReceived</i>	Control	14258	26.178	52.207		
	<i>MessagesSent</i>	Treatment	14291	23.860	72.206	-0.9243	0.3553
	<i>MessagesSent</i>	Control	14258	24.694	80.000		
	<i>MessagesReceived</i>	Treatment	14291	5.580	13.238	-1.4408	0.1497
	<i>MessagesReceived</i>	Control	14258	5.816	14.461		
	<i>MatchesSent</i>	Treatment	14291	2.518	7.761	-1.1478	0.2511
	<i>MatchesSent</i>	Control	14258	2.628	8.455		
	<i>MatchesReceived</i>	Treatment	14291	0.423	1.248	-1.1048	0.2692
	<i>MatchesReceived</i>	Control	14258	0.439	1.283		

Table 2 – Descriptive Statistics for Individual Characteristics

Gender	Basic Outcomes	Manipulation	Observations	Mean	Std. Error	t-statistic	p-value
Females	Age	Treatment	5614	33.026	11.508	-0.8399	0.4010
	Age	Control	5557	33.211	11.746		
	White	Treatment	5614	0.234	0.423	-0.0970	0.9227
	White	Control	5557	0.235	0.424		
	Desirability Score	Treatment	5614	50.392	72.994	0.5088	0.6109
	Desirability Score	Control	5557	49.686	73.782		
Males	Age	Treatment	14291	30.578	10.086	-1.1542	0.2484
	Age	Control	14258	30.717	10.156		
	White	Treatment	14291	0.228	0.420	-1.1464	0.2516
	White	Control	14258	0.234	0.423		
	Desirability Score	Treatment	14291	5.580	13.238	-1.4408	0.1497
	Desirability Score	Control	14258	5.816	14.461		

Table 3 – Descriptive Statistics for Basic Outcomes

Gender	Basic Outcomes	Manipulation	Observations	Mean	Std. Error	t- statistic	p- value
Females	<i>ViewsSent</i>	Treatment	5614	30.469	80.729	1.7747	0.0760
	<i>ViewsSent</i>	Control	5557	26.994	121.898		
	<i>ViewsReceived</i>	Treatment	5614	97.649	156.515	1.2184	0.2231
	<i>ViewsReceived</i>	Control	5557	93.827	174.488		
	<i>MessagesSent</i>	Treatment	5614	6.229	15.156	1.8060	0.0709
	<i>MessagesSent</i>	Control	5557	5.711	15.111		
	<i>MessagesReceived</i>	Treatment	5614	22.842	45.284	1.0580	0.2901
	<i>MessagesReceived</i>	Control	5557	21.913	47.511		
	<i>MatchesSent</i>	Treatment	5614	0.496	2.141	1.9130	0.0558
	<i>MatchesSent</i>	Control	5557	0.425	1.800		
	<i>MatchesReceived</i>	Treatment	5614	2.620	6.911	2.9835	0.0029
	<i>MatchesReceived</i>	Control	5557	2.251	6.150		
Males	<i>ViewsSent</i>	Treatment	14291	50.808	146.720	0.1293	0.8971
	<i>ViewsSent</i>	Control	14258	50.578	152.543		
	<i>ViewsReceived</i>	Treatment	14291	12.630	32.923	-0.0803	0.9360
	<i>ViewsReceived</i>	Control	14258	12.661	32.278		
	<i>MessagesSent</i>	Treatment	14291	12.538	50.691	0.7552	0.4501
	<i>MessagesSent</i>	Control	14258	12.096	48.136		
	<i>MessagesReceived</i>	Treatment	14291	3.227	10.264	1.2128	0.2252
	<i>MessagesReceived</i>	Control	14258	3.084	9.665		
	<i>MatchesSent</i>	Treatment	14291	1.247	5.225	1.4396	0.1500
	<i>MatchesSent</i>	Control	14258	1.159	5.071		
	<i>MatchesReceived</i>	Treatment	14291	0.204	0.899	1.6599	0.0970
	<i>MatchesReceived</i>	Control	14258	0.187	0.769		

Table 4 – Female: NB Model for Basic Outcomes

VARIABLES	(1) <i>ViewsReceived</i>	(2) <i>ViewsSent</i>	(3) <i>MessagesReceived</i>	(4) <i>MessagesSent</i>	(5) <i>MatchesReceived</i>	(6) <i>MatchesSent</i>
Treatment	0.0507** (0.0253)	0.0989** (0.0413)	0.0548** (0.0276)	0.0711* (0.0399)	0.1335*** (0.0429)	0.1346** (0.0604)
Control Variables						
Age	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓	✓	✓
Pre-treatment activities (PC1 only)	✓	✓	✓	✓	✓	✓
Constant Term	4.1766*** (0.1015)	1.9828*** (0.1654)	2.6288*** (0.1102)	0.9811*** (0.1573)	0.1468 (0.1670)	-1.7360*** (0.2261)
Observations	11171	11171	11171	11171	11171	11171

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 5 – Male: NB Model for Basic Outcomes

VARIABLES	(1) <i>ViewsReceived</i>	(2) <i>ViewsSent</i>	(3) <i>MessagesReceived</i>	(4) <i>MessagesSent</i>	(5) <i>MatchesReceived</i>	(6) <i>MatchesSent</i>
Treatment	0.0333* (0.0182)	0.0482* (0.0278)	0.0700*** (0.0231)	0.0855*** (0.0303)	0.1362*** (0.0414)	0.1089*** (0.0332)
Control Variables						
Age	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓	✓	✓
Pre-treatment	✓	✓	✓	✓	✓	✓
activities (PC1 only)						
Constant Term	1.0217*** (0.0751)	2.9366*** (0.1155)	-0.0952 (0.0945)	1.7145*** (0.1256)	-2.7514*** (0.1639)	-0.7674*** (0.1358)
Observations	28549	28549	28549	28549	28549	28549

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 6 – Descriptive Statistics for Advanced Outcomes

Gender	Advanced Outcomes	Manipulation	Observations	Mean	Std. Error	t- statistic	p- value
Females	<i>VoteViewsSent</i>	Treatment	5614	30.234	80.759	1.7647	0.0776
	<i>VoteViewsSent</i>	Control	5557	26.778	121.889		
	<i>VoteViewMessagesReceived</i>	Treatment	5614	5.415	12.642	1.1062	0.2687
	<i>VoteViewMessagesReceived</i>	Control	5557	5.019	23.546		
Males	<i>VoteViewsSent</i>	Treatment	14291	48.960	145.842	0.2347	0.8144
	<i>VoteViewsSent</i>	Control	14258	48.547	151.556		
	<i>VoteViewMessagesReceived</i>	Treatment	14291	2.366	8.719	1.2400	0.2150
	<i>VoteViewMessagesReceived</i>	Control	14258	2.240	8.451		

Table 7 – Female: NB Model for Advanced Outcomes

VARIABLES	(1) <i>VoteViewsSent</i>	(2) <i>VoteViewMessagesReceived</i>
Treatment	0.0982** (0.0425)	0.1095*** (0.0387)
Control Variables		
Age	✓	✓
Body Types	✓	✓
Education Levels	✓	✓
Ethnicities	✓	✓
Pre-treatment	✓	✓
activities (PC1 only)		
Constant Term	1.9945*** (0.1702)	0.5357*** (0.1530)
Observations	11171	11171

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 8 – Male: NB Model for Advanced Outcomes

VARIABLES	(1) <i>VoteViewsSent</i>	(2) <i>VoteViewMessagesReceived</i>
Treatment	0.0685** (0.0299)	0.0859*** (0.0283)
Control Variables		
Age	✓	✓
Body Types	✓	✓
Education Levels	✓	✓
Ethnicities	✓	✓
Pre-treatment	✓	✓
activities (PC1 only)		
Constant Term	3.0248*** (0.1243)	-0.3287*** (0.1160)
Observations	28549	28549

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 9 – Female: Match Efficiency and Time to Match Outcomes

VARIABLES	(1)	(2)	(3)	(4)
	$\frac{MatchesReceived}{MessagesReceived}$	$\frac{MatchesSent}{MessagesSent}$	Time to first match received	Time to first match sent
Treatment	0.0076*** (0.0027)	0.0056 (0.0037)	0.1106** (0.0340)	0.0503 (0.0518)
Control Variables				
Age	✓	✓	✓	✓
Body Types	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓
Pre-treatment activities	✓	✓	✓	✓
(PC1 only)				
Constant Term	0.0667*** (0.0105)	0.0250* (0.0136)		
Observations	8,462	4,889	10,218	10,218

Note. Models (1) and (2) uses ordinary least square regression, while Models (3) and (4) uses Cox-Proportional Hazard model. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 10 – Male: Match Efficiency and Time to Match Outcomes

VARIABLES	(1)	(2)	(3)	(4)
	$\frac{MatchesReceived}{MessagesReceived}$	$\frac{MatchesSent}{MessagesSent}$	Time to first match received	Time to first match sent
Treatment	0.0034 (0.0032)	0.0020 (0.0031)	0.1348*** (0.0385)	0.0940*** (0.0270)
Control Variables				
Age	✓	✓	✓	✓
Body Types	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓
Pre-treatment activities	✓	✓	✓	✓
(PC1 only)				
Constant Term	0.0839*** (0.0124)	0.0951*** (0.0121)		
Observations	11,682	11,522	24,115	24,115

Note. Models (1) and (2) uses ordinary least square regression, while Models (3) and (4) uses Cox-Proportional Hazard model. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 11 – Female: NB Model for Heterogeneity Analysis by Desirability

VARIABLES		(1) <i>MatchesReceived</i>	(2) <i>MatchesSent</i>
(1)	Treatment	0.4445*** (0.1593)	0.8596*** (0.2679)
(2)	Focal User's Desirability (High)	0.9863*** (0.1446)	0.6383*** (0.2446)
(3)	Focal User's Desirability (Medium)	0.9571*** (0.1465)	1.0598*** (0.2495)
(4)	Targeting Users' Desirability (High)	-0.0323 (0.1570)	0.9079*** (0.2541)
(5)	Targeting Users' Desirability (Medium)	0.3680** (0.1707)	0.2347 (0.3106)
(6)	Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	0.1955 (0.1979)	-0.5590* (0.3014)
(7)	Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	-0.0844 (0.2000)	-1.0592*** (0.3109)
(8)	Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	-0.3072 (0.2041)	-0.1971 (0.3471)
(9)	Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	-0.3570* (0.2103)	-0.3540 (0.3591)
(10)	Treatment ★ Focal User's Desirability (High)	-0.3125 (0.1951)	-0.6964** (0.3103)
(11)	Treatment ★ Focal User's Desirability (Medium)	-0.6336*** (0.2042)	-1.1781*** (0.3306)
(12)	Treatment ★ Targeting Users' Desirability (High)	0.1807 (0.2144)	-1.0077*** (0.3379)
(13)	Treatment ★ Targeting Users' Desirability (Medium)	-0.5100** (0.2402)	-0.4829 (0.4059)
(14)	Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	-0.3565 (0.2722)	0.8573** (0.4054)
(15)	Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	0.2333 (0.2762)	1.4497*** (0.4237)
(16)	Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	0.4351 (0.2863)	0.4917 (0.4588)
(17)	Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	0.9298*** (0.2958)	1.1484** (0.4798)
(18)	Constant Term	-0.7576*** (0.1985)	-2.5658*** (0.3079)
Control Variables –			
Age, Body Types, Education Levels, Ethnicities		✓	✓
Pre-treatment activities (PC1 only)		✓	✓
Observations		10218	10218

Robust standard errors in parentheses
 *** p < 0.01, ** p < 0.05, * p < 0.1

Table 12 – Male: NB Model for Heterogeneity Analysis by Desirability

VARIABLES		(1) <i>MatchesReceived</i>	(2) <i>MatchesSent</i>
(1)	Treatment	0.5404** (0.2298)	0.1268 (0.1253)
(2)	Focal User's Desirability (High)	1.8997*** (0.1969)	1.7646*** (0.1110)
(3)	Focal User's Desirability (Medium)	1.0775*** (0.2074)	0.5340*** (0.1149)
(4)	Targeting Users' Desirability (High)	0.0052 (0.2657)	-0.5728*** (0.1495)
(5)	Targeting Users' Desirability (Medium)	0.3275 (0.2606)	-0.2457* (0.1493)
(6)	Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	0.1586 (0.2814)	0.5477*** (0.1698)
(7)	Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	-0.2891 (0.3059)	0.4725*** (0.1810)
(8)	Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	-0.1373 (0.2750)	0.1192 (0.1681)
(9)	Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	-0.2175 (0.2960)	0.2939 (0.1804)
(10)	Treatment ★ Focal User's Desirability (High)	-0.3642 (0.2499)	-0.0233 (0.1515)
(11)	Treatment ★ Focal User's Desirability (Medium)	-0.5541** (0.2682)	-0.1805 (0.1591)
(12)	Treatment ★ Targeting Users' Desirability (High)	-0.3053 (0.3464)	0.3062 (0.2001)
(13)	Treatment ★ Targeting Users' Desirability (Medium)	-0.3631 (0.3434)	-0.0339 (0.2075)
(14)	Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	0.1424 (0.3695)	-0.3977* (0.2300)
(15)	Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	0.5498 (0.4042)	0.0129 (0.2453)
(16)	Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	0.3959 (0.3644)	0.0820 (0.2344)
(17)	Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	0.3359 (0.3970)	0.0916 (0.2521)
(18)	Constant Term	-3.9797*** (0.2378)	-1.6441*** (0.1556)
Control Variables –			
Age, Body Types, Education Levels, Ethnicities		✓	✓
Pre-treatment activities (PC1 only)		✓	✓
Observations		24115	24115

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 13 – Female: NB Model for Robustness Related Outcomes

TIME FRAMES	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Monthly Level		Weekly Level				
	Month 2	Month 3	Month 2 Week 1	Month 2 Week 2	Month 2 Week 3	Month 2 Week 4	Month 3 Week 1
Treatment	0.1335*** (0.0429)	0.0357 (0.0514)	0.1660*** (0.0495)	0.1033* (0.0534)	0.0872 (0.0557)	0.1371** (0.0577)	-0.0473 (0.0599)
Control Variables							
Age	✓	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓	✓	✓	✓
Pre-treatment activities (PC1 only)	✓	✓	✓	✓	✓	✓	✓
Constant Term	0.1470 (0.1670)	0.0942 (0.2011)	- 1.0754*** (0.1859)	- 1.0554*** (0.2015)	- 1.0586*** (0.2110)	- 1.3570*** (0.2194)	- 1.1594*** (0.2282)
Observations	11171	11171	11171	11171	11171	11171	11171

Notes. Month 2 is the experiment month, and Month 3 is the post-experiment month. The dependent variable for columns (1) to (7) is MatchesReceived. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 14 – Male: NB Model for Robustness Related Outcomes

TIME FRAMES	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Monthly Level		Weekly Level				
	Month 2	Month 3	Month 2 Week 1	Month 2 Week 2	Month 2 Week 3	Month 2 Week 4	Month 3 Week 1
Treatment	0.1358*** (0.0414)	0.0279 (0.0466)	0.1403** (0.0637)	0.1837*** (0.0672)	0.1810** (0.0711)	0.0147 (0.0725)	0.0787 (0.0740)
Control Variables							
Age	✓	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓	✓	✓	✓
Pre-treatment activities (PC1 only)	✓	✓	✓	✓	✓	✓	✓
Constant Term	-	-	-	-	-	-	-
	2.7521*** (0.1639)	2.9101*** (0.1817)	4.1264*** (0.2564)	4.0872*** (0.2663)	4.4103*** (0.2858)	3.7788*** (0.2766)	4.0760*** (0.2833)
Observations	28549	28549	28549	28549	28549	28549	28549

Notes. Month 2 is the experiment month, and Month 3 is the post-experiment month. The dependent variable for columns (1) to (7) is MatchesReceived. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Appendix A

Table A1 – Model Selection Using Akaike information criterion (AIC)

AIC Calculation Across Models	(1) <i>MatchesReceived</i>	(2) <i>MatchesSent</i>
(1) Female only NB Model for heterogeneity analysis by desirability		
No. of Observations	10218	10218
No. of parameters	45	45
Log Likelihood	-16378.5558	-6996.1094
AIC	32849.1117	14084.2187
(2) Male only NB Model for heterogeneity analysis by desirability		
No. of Observations	24115	24115
No. of parameters	45	45
Log Likelihood	-11252.3608	-24929.5167
AIC	22596.7215	49951.0335
(3) Combined AIC from (1) and (2)	55441.8332	64031.2522
(4) Pooled NB Model for heterogeneity analysis by desirability		
No. of Observations	34333	34333
No. of parameters	90	90
Log Likelihood	-27658.4791	-31947.6538
AIC	55498.9583	64077.3076

Note: Combined AIC in (3) = $-2 * (\log \text{likelihood in (1)} + \log \text{likelihood in (2)}) + 2 (\text{no. of parameter in (1)} + \text{no. of parameter in (2)})$.

Appendix B: SUTVA Corrected Models

Table B1: Neighbors' Exposure to Treatment in the *ViewsSentPre* Network

For Treated Focal Females (Sample size = 11,171) –

	Total No. of Neighbors	No. of Treated Neighbors	Percentage of Treated Neighbors = $\left(\frac{\text{No. of Treated Neighbors}}{\text{Total No. of Neighbors}} * 100 \right)$
Mean	55.25	0.37	1.01
Std.	109.13	0.87	5.21
Median	21	0	0

For Treated Focal Males (Sample size = 28,549) –

	Total No. of Neighbors	No. of Treated Neighbors	Percentage of Treated Neighbors = $\left(\frac{\text{No. of Treated Neighbors}}{\text{Total No. of Neighbors}} * 100 \right)$
Mean	89.69	0.50	0.55
Std	197.74	0.50	1.36
Median	26	1	0

Next, we use the more flexible regression adjustment following Chin (2019) to tackle the second order SUTVA violation scenario, where the treated and control group users are connected via a third person, with whom they each interact directly.

To calculate the correction term we:

1. **Construct the underlying directed networks for user engagement and matching behavior for the pre-treatment month.** To understand this network, let's consider the 'matches sent' network. Here, the nodes represent dating platform users, which include focal users (i.e., users who eventually become part of the experiment during the experiment month) and other users to whom focal users sent a match. Any directed edge in this network, say from node A to node B, represents a match sent from node A to node B.
2. **Calculate the correction term for each focal user.** In the matches sent network, the correction term is the proportion of the focal user's neighbors who have received matches from at least another user from a target cohort. Note that the target cohort of users depend on the focal user's cohort. For example, if the focal user belongs to the treated female cohort, then the target users belong to the control female cohort. In other words, at least one individual from the target cohort and the focal user have both sent matches to a common neighbor. We term such a neighbor as an infected neighbor. Therefore, the correction term is the total number of infected neighbors divided by the total number of neighbors for a focal user. Further, the correction term calculation here is the proportion of the treated female focal user's neighbors who have received matches from at least one control group female user, in the pre-treatment period. Mathematically:

$$SUTVA \text{ Violation Correction Term}_i = \frac{\sum_j \text{Infected Neighbor}_{ij}}{\sum_j \text{Neighbor}_{ij}}$$

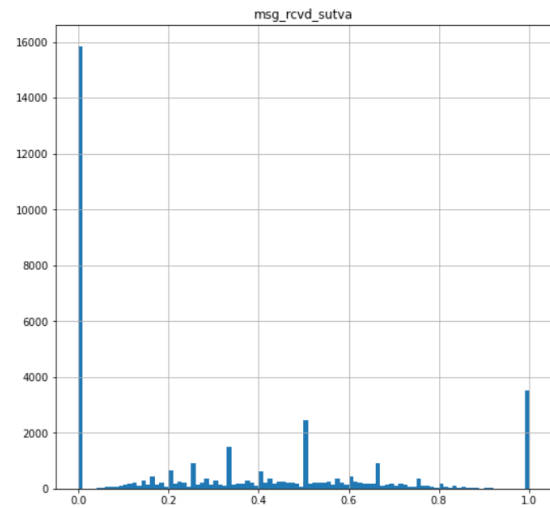
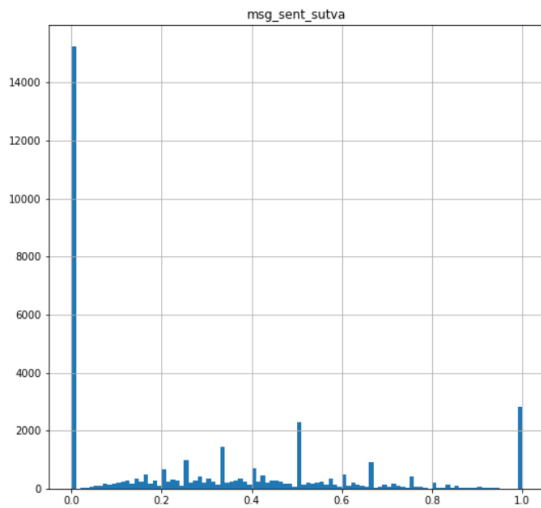
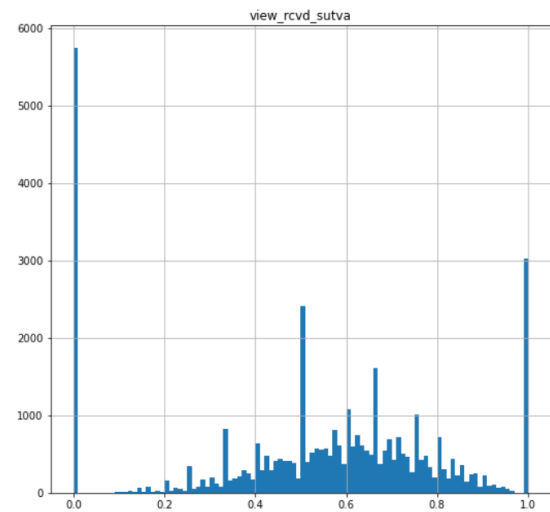
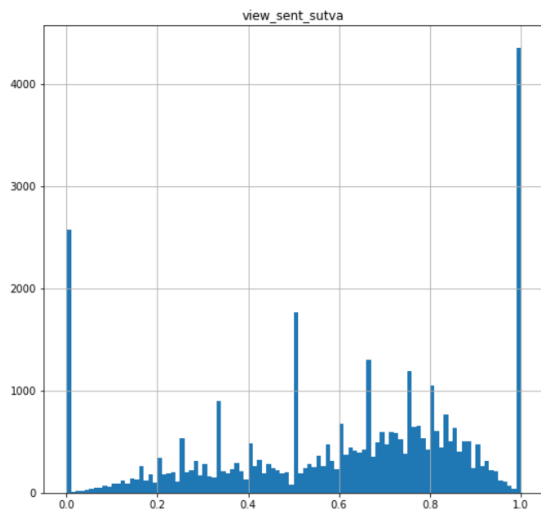
$$\text{Neighbor}_{ij} = 1 \text{ if node } j \text{ is a neighbor of node } i;$$

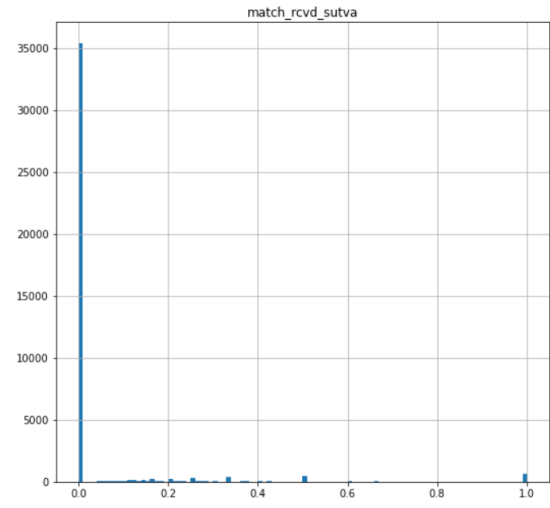
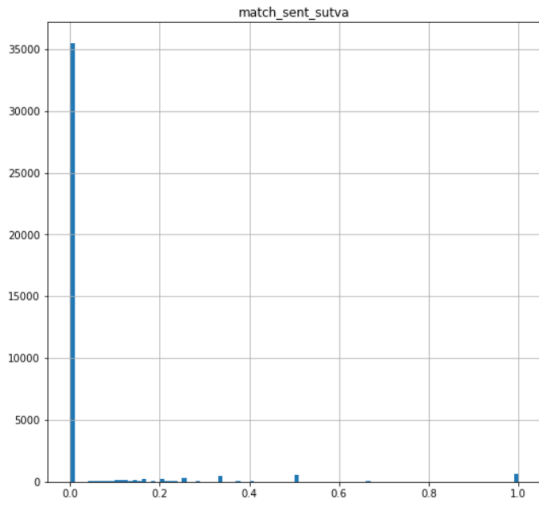
$$0 \text{ otherwise}$$

Infected Neighbor_{ij}

= 1 if node j is a neighbor of i and node j's neighbors contain at least one user from a target cohort;
0 otherwise

3. Create histograms of the SUTVA violation correction term. Here, we present histograms for the frequency distribution of the SUTVA violation correction term across six networks – views sent, views received, messages sent, messages received, matches sent, and matches received. Here, the x-axis is the value of the correction term (which is between 0 and 1), and the y-axis is the frequency (i.e., number of focal users).





4. **Add correction terms in the regression models (as suggested by Chin 2019).** For example, if the dependent variable is match sent, then we use the corresponding correction term from the match sent network. Similarly, for other dependent variables, we use individual or multiple correction terms based on the networks involved for those analyses. This way, we deal with SUTVA violation in different contexts uniquely. For instance, the above figure clearly shows that SUTVA violation is very evident for views sent and views received networks, yet not that prominent for matches sent and matches received networks. So, we utilize the correction terms in our regressions as the context demands.

Table B2 – Female: SUTVA Corrected NB Model for Basic Outcomes

VARIABLES	(1) <i>ViewsReceived</i>	(2) <i>ViewsSent</i>	(3) <i>MessagesReceived</i>	(4) <i>MessagesSent</i>	(5) <i>MatchesReceived</i>	(6) <i>MatchesSent</i>
Treatment	0.0452* (0.0250)	0.0978** (0.0413)	0.0444 (0.0274)	0.0635 (0.0397)	0.1414*** (0.0428)	0.1305** (0.0603)
Control Variables						
Age	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓
Education	✓	✓	✓	✓	✓	✓
Levels						
Ethnicities	✓	✓	✓	✓	✓	✓
Pre-treatment activities (PC1 only)	✓	✓	✓	✓	✓	✓
SUTVA Correction Term	1.4994*** (0.0599)	0.0527 (0.0876)	1.0942*** (0.0538)	0.9088*** (0.0864)	0.9691*** (0.1006)	0.9727** (0.3834)
Constant Term	3.2403*** (0.1077)	1.1976*** (0.1675)	2.0788*** (0.1127)	0.7910*** (0.1577)	-0.0354 (0.1673)	-1.7454*** (0.2259)
Observations	11171	11171	11171	11171	11171	11171

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B3 – Male: SUTVA Corrected NB Model for Basic Outcomes

VARIABLES	(1) <i>ViewsReceived</i>	(2) <i>ViewsSent</i>	(3) <i>MessagesReceived</i>	(4) <i>MessagesSent</i>	(5) <i>MatchesReceived</i>	(6) <i>MatchesSent</i>
Treatment	0.0265 (0.0180)	0.0436 (0.0278)	0.0718*** (0.0230)	0.0989*** (0.0299)	0.1348*** (0.0414)	0.1101*** (0.0332)
Control Variables						
Age	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓
Education	✓	✓	✓	✓	✓	✓
Levels						
Ethnicities	✓	✓	✓	✓	✓	✓
Pre-treatment activities (PC1 only)	✓	✓	✓	✓	✓	✓
SUTVA Correction Term	1.0801*** (0.0311)	0.5002*** (0.0634)	0.2913*** (0.0362)	1.4540*** (0.0468)	0.5680*** (0.1396)	0.8573*** (0.0888)
Constant Term	0.3205*** (0.0774)	2.5034*** (0.1277)	-0.1766* (0.0950)	0.8983*** (0.1265)	-2.7707*** (0.1639)	-0.8732*** (0.1359)
Observations	28549	28549	28549	28549	28549	28549

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B4 – Female: SUTVA Corrected NB Model for Advanced Outcomes

VARIABLES	(1) <i>VoteViewsSent</i>	(2) <i>VoteViewMessagesReceived</i>
Treatment	0.0944** (0.0425)	0.0939** (0.0386)
Control Variables		
Age	✓	✓
Body Types	✓	✓
Education Levels	✓	✓
Ethnicities	✓	✓
Pre-treatment activities	✓	✓
(PC1 only)		
SUTVA Correction Terms	✓	✓
Constant Term	1.9474*** (0.1723)	0.0881 (0.1580)
Observations	11171	11171

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B5 – Male: SUTVA Corrected NB Model for Advanced Outcomes

VARIABLES	(1) <i>VoteViewsSent</i>	(2) <i>VoteViewMessagesReceived</i>
Treatment	0.0634** (0.0299)	0.0884*** (0.0283)
Control Variables		
Age	✓	✓
Body Types	✓	✓
Education Levels	✓	✓
Ethnicities	✓	✓
Pre-treatment activities	✓	✓
(PC1 only)		
SUTVA Correction Terms	✓	✓
Constant Term	2.5606*** (0.1375)	-0.4838*** (0.1303)
Observations	28549	28549

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B6 – Female: SUTVA Corrected NB Models for Match Efficiency and Time to Match Outcomes

VARIABLES	(1)	(2)	(3)	(4)
	$\frac{MatchesReceived}{MessagesReceived}$	$\frac{MatchesSent}{MessagesSent}$	Time to first match received	Time to first match sent
Treatment	0.0079*** (0.0027)	0.0055 (0.0037)	0.1138*** (0.0340)	0.0495 (0.0519)
Control Variables				
Age	✓	✓	✓	✓
Body Types	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓
Pre-treatment activities	✓	✓	✓	✓
(PC1 only)				
SUTVA Correction Terms	✓	✓	✓	✓
Constant Term	0.0647*** (0.0110)	0.0319** (0.0137)		
Observations	8,462	4,889	10,218	10,218

Note. Models (1) and (2) uses ordinary least square regression, while Models (3) and (4) uses Cox-Proportional Hazard model. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B7 – Male: SUTVA Corrected NB Models for Match Efficiency and Time to Match Outcomes

VARIABLES	(1)	(2)	(3)	(4)
	$\frac{MatchesReceived}{MessagesReceived}$	$\frac{MatchesSent}{MessagesSent}$	Time to first match received	Time to first match sent
Treatment	0.0034 (0.0032)	0.0020 (0.0031)	0.1356*** (0.0385)	0.0902*** (0.0270)
Control Variables				
Age	✓	✓	✓	✓
Body Types	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓
Pre-treatment activities	✓	✓	✓	✓
(PC1 only)				
SUTVA Correction Terms	✓	✓	✓	✓
Constant Term	0.0823*** (0.0126)	0.1035*** (0.0127)		
Observations	11,682	11,522	24,115	24,115

Note. Models (1) and (2) uses ordinary least square regression, while Models (3) and (4) uses Cox-Proportional Hazard model. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B8 – Female: SUTVA Corrected NB Model for Heterogeneity Analysis by Desirability

VARIABLES	(1) <i>MatchesReceived</i>	(2) <i>MatchesSent</i>
(1) Treatment	0.4638*** (0.1595)	0.8603*** (0.2676)
(2) Focal User's Desirability (High)	0.9804*** (0.1449)	0.6419*** (0.2444)
(3) Focal User's Desirability (Medium)	0.9480*** (0.1468)	1.0644*** (0.2493)
(4) Targeting Users' Desirability (High)	-0.0460 (0.1575)	0.9104*** (0.2539)
(5) Targeting Users' Desirability (Medium)	0.3906** (0.1709)	0.2310 (0.3106)
(6) Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	0.1415 (0.1981)	-0.5636* (0.3011)
(7) Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	-0.1151 (0.2003)	-1.0687*** (0.3107)
(8) Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	-0.3489* (0.2041)	-0.1956 (0.3471)
(9) Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	-0.4110* (0.2103)	-0.3562 (0.3590)
(10) Treatment ★ Focal User's Desirability (High)	-0.3262* (0.1951)	-0.6954** (0.3099)
(11) Treatment ★ Focal User's Desirability (Medium)	-0.6584*** (0.2043)	-1.1806*** (0.3303)
(12) Treatment ★ Targeting Users' Desirability (High)	0.1888 (0.2146)	-1.0081*** (0.3376)
(13) Treatment ★ Targeting Users' Desirability (Medium)	-0.5215** (0.2400)	-0.4754 (0.4057)
(14) Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	-0.3530 (0.2721)	0.8454** (0.4049)
(15) Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	0.2132 (0.2762)	1.4357*** (0.4234)
(16) Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	0.4514 (0.2859)	0.4811 (0.4585)
(17) Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	0.9497*** (0.2956)	1.1442** (0.4795)
(18) Constant Term	-0.8471*** (0.1989)	-2.5765*** (0.3077)
Control Variables –		
Age, Body Types, Education Levels, Ethnicities	✓	✓
Pre-treatment activities (PC1 only)	✓	✓
SUTVA correction term	✓	✓
Observations	10218	10218

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B9 – Male: SUTVA Corrected NB Model for Heterogeneity Analysis by Desirability

VARIABLES	(1) <i>MatchesReceived</i>	(2) <i>MatchesSent</i>
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(1)	Treatment	0.5404** (0.2298)	0.1270 (0.1253)
(2)	Focal User's Desirability (High)	1.8963*** (0.1969)	1.7529*** (0.1114)
(3)	Focal User's Desirability (Medium)	1.0765*** (0.2074)	0.5300*** (0.1149)
(4)	Targeting Users' Desirability (High)	0.0051 (0.2657)	-0.5721*** (0.1495)
(5)	Targeting Users' Desirability (Medium)	0.3274 (0.2606)	-0.2453 (0.1494)
(6)	Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	0.1545 (0.2814)	0.5436*** (0.1699)
(7)	Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	-0.2896 (0.3059)	0.4679*** (0.1810)
(8)	Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	-0.1401 (0.2750)	0.1171 (0.1681)
(9)	Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	-0.2187 (0.2960)	0.2941 (0.1804)
(10)	Treatment ★ Focal User's Desirability (High)	-0.3644 (0.2498)	-0.0233 (0.1516)
(11)	Treatment ★ Focal User's Desirability (Medium)	-0.5543** (0.2682)	-0.1806 (0.1591)
(12)	Treatment ★ Targeting Users' Desirability (High)	-0.3052 (0.3464)	0.3059 (0.2001)
(13)	Treatment ★ Targeting Users' Desirability (Medium)	-0.3631 (0.3433)	-0.0336 (0.2075)
(14)	Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (High)	0.1433 (0.3694)	-0.3972* (0.2300)
(15)	Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (High)	0.5488 (0.4042)	0.0125 (0.2453)
(16)	Treatment ★ Focal User's Desirability (High) ★ Targeting Users' Desirability (Medium)	0.3958 (0.3643)	0.0811 (0.2344)
(17)	Treatment ★ Focal User's Desirability (Medium) ★ Targeting Users' Desirability (Medium)	0.3356 (0.3970)	0.0901 (0.2521)
(18)	Constant Term	-3.9816*** (0.2378)	-1.6514*** (0.1557)
Control Variables –			
	Age, Body Types, Education Levels, Ethnicities	✓	✓
	Pre-treatment activities (PC1 only)	✓	✓
	SUTVA Correction term	✓	✓
Observations		24115	24115

Robust standard errors in parentheses

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B10 – Female: SUTVA Corrected NB Model for Robustness Related Outcomes

TIME FRAMES	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Monthly Level		Weekly Level				
	Month 2	Month 3	Month 2 Week 1	Month 2 Week 2	Month 2 Week 3	Month 2 Week 4	Month 3 Week 1
Treatment	0.1414*** (0.0428)	0.0428 (0.0513)	0.1664*** (0.0494)	0.1100** (0.0533)	0.0938* (0.0556)	0.1477** (0.0577)	-0.0327 (0.0598)
Control Variables							
Age	✓	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓	✓	✓	✓
Pre-treatment activities	✓	✓	✓	✓	✓	✓	✓
(PC1 only)							
SUTVA Correction Term	0.9691*** (0.1006)	1.0017*** (0.1215)	1.1049*** (0.1104)	0.8729*** (0.1218)	0.7851*** (0.1281)	0.8407*** (0.1324)	0.9689*** (0.1362)
Constant Term	-0.0354 (0.1673)	-0.0827 (0.2016)	-1.3002*** (0.1866)	-1.2084*** (0.2024)	-1.1969*** (0.2118)	-1.5149*** (0.2205)	-1.3720*** (0.2293)
Observations	11171	11171	11171	11171	11171	11171	11171

Notes. Month 2 is the experiment month, and Month 3 is the post-experiment month. The dependent variable for columns (1) to (7) is MatchesReceived. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table B11 – Male: SUTVA Corrected NB Model for Robustness Related Outcomes

TIME FRAMES	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Monthly Level		Weekly Level				
	Month 2	Month 3	Month 2 Week 1	Month 2 Week 2	Month 2 Week 3	Month 2 Week 4	Month 3 Week 1
Treatment	0.1348*** (0.0414)	0.0290 (0.0465)	0.1403** (0.0636)	0.1820*** (0.0671)	0.1831*** (0.0710)	0.0123 (0.0724)	0.0789 (0.0739)
Control Variables							
Age	✓	✓	✓	✓	✓	✓	✓
Body Types	✓	✓	✓	✓	✓	✓	✓
Education Levels	✓	✓	✓	✓	✓	✓	✓
Ethnicities	✓	✓	✓	✓	✓	✓	✓
Pre-treatment activities	✓	✓	✓	✓	✓	✓	✓
(PC1 only)							
SUTVA Correction Term	0.5680*** (0.1396)	0.6231*** (0.1566)	0.6236*** (0.2015)	0.7200*** (0.2065)	0.5947*** (0.2271)	0.5790** (0.2323)	0.5575** (0.2384)
Constant Term	-2.7707*** (0.1639)	-2.9239*** (0.1816)	-4.1480*** (0.2565)	-4.1010*** (0.2659)	-4.4223*** (0.2853)	-3.8127*** (0.2772)	-4.0813*** (0.2829)
Observations	28549	28549	28549	28549	28549	28549	28549

Notes. Month 2 is the experiment month, and Month 3 is the post-experiment month. The dependent variable for columns (1) to (7) is MatchesReceived. Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1